

Designing Accessible Apps



Intro

Summary

- Intro
- People with disabilities & digital difficulties
- Accessible Apps
 - Layout
 - Content
 - Environmental
- Make it sexy
- Needed apps
- Conclusion
- Thanks || Q&A



Who am I?

- Dennie Declercq
- Volunteer developer!
- Microsoft MVP
- DDSOft (nonprofit)
- #DreamingIsBelieving
- Positive thinker





#7ThYearMagic



People with disabilities

& digital difficulties

People!



- PWD are People
- People
- Thread human
- Help Human

Principles of Inclusion

- Same activities
- Same district
- Work Together
- Adjustments par person



Kind of disabilities



Intellectual
disabilities



Autism



Visual



Auditory



Motoric

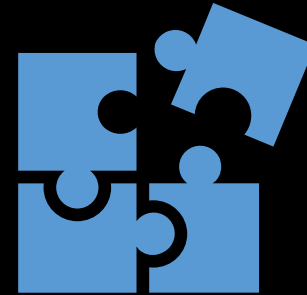


Much more

Focus

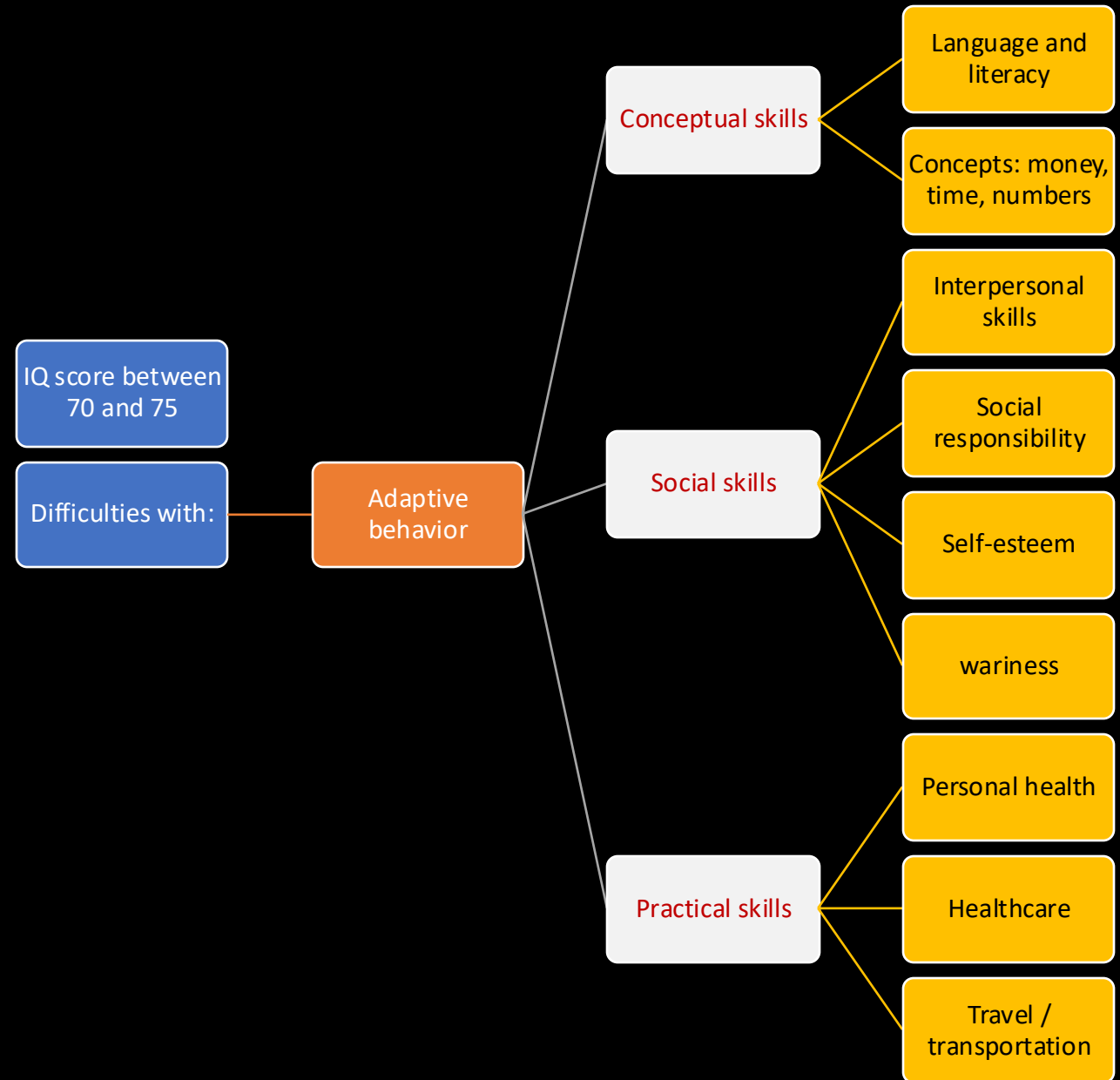


Cognitive / intellectual / learning
disabilities



Autism

Focus: Cognitive / intellectual / learning disabilities

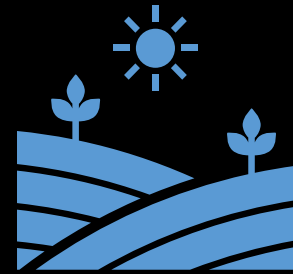


Focus: autism



Mismatch in:

Communication
Executive functioning
Central coherence

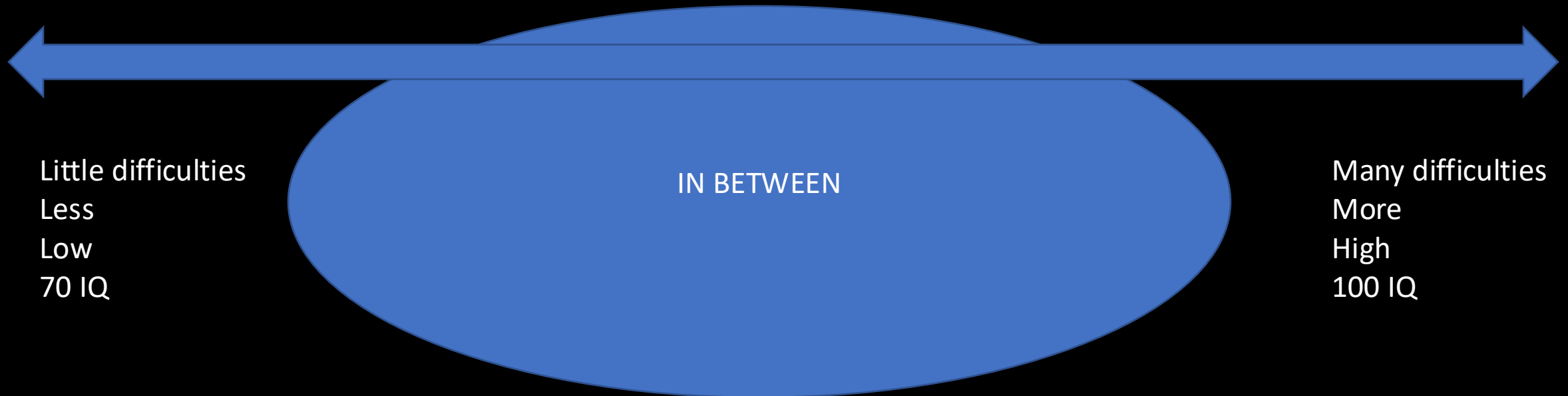


Difficulties

All fields
Every situation in life

Digital difficulties

- A spectrum



Digital difficulties / Cognitive Disability

- Understand
 - What to do with application?
 - Which controls do I have to press?
 - How to navigate?
 - Math , counting
 - Structure
 - Screen
 - Whole app
 - Layout
 - Stack instead list

Digital difficulties / Autism



Structure



Which buttons do I have to press?



Concentration



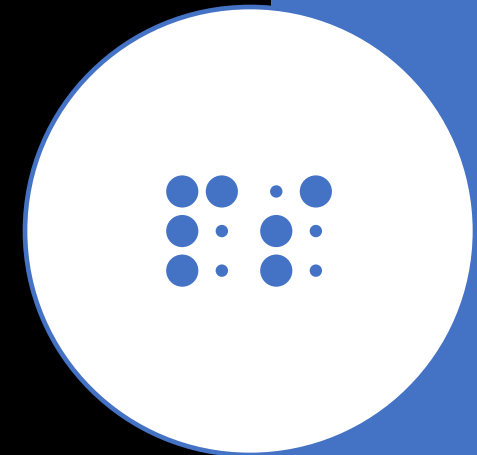
Also accessible for:

ADHD

ADD

Digital difficulties / Colorblind

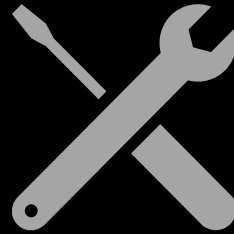
- Troubles when
 - Non-matching color scheme
 - Most famous: Red-Green
- BUT
 - Colors == really helpful for:
 - Intellectual disability
 - Autism
 - Red-Green
 - Really clear for:
 - Learning disabilities / intellectual disability
 - Autism



Digital difficulties / Low vision



Seeing screen /
What's on the screen?

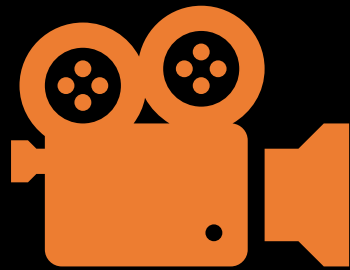


Use zooming tools

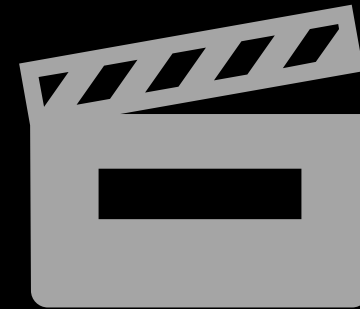


Use speech tools

Digital Difficulties / Hearing difficulties



Video's



Introduction video's

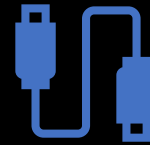
Digital Difficulties / Paralyzed



Hand control



Mouse



Device

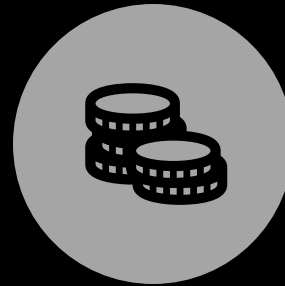


Multi-touch

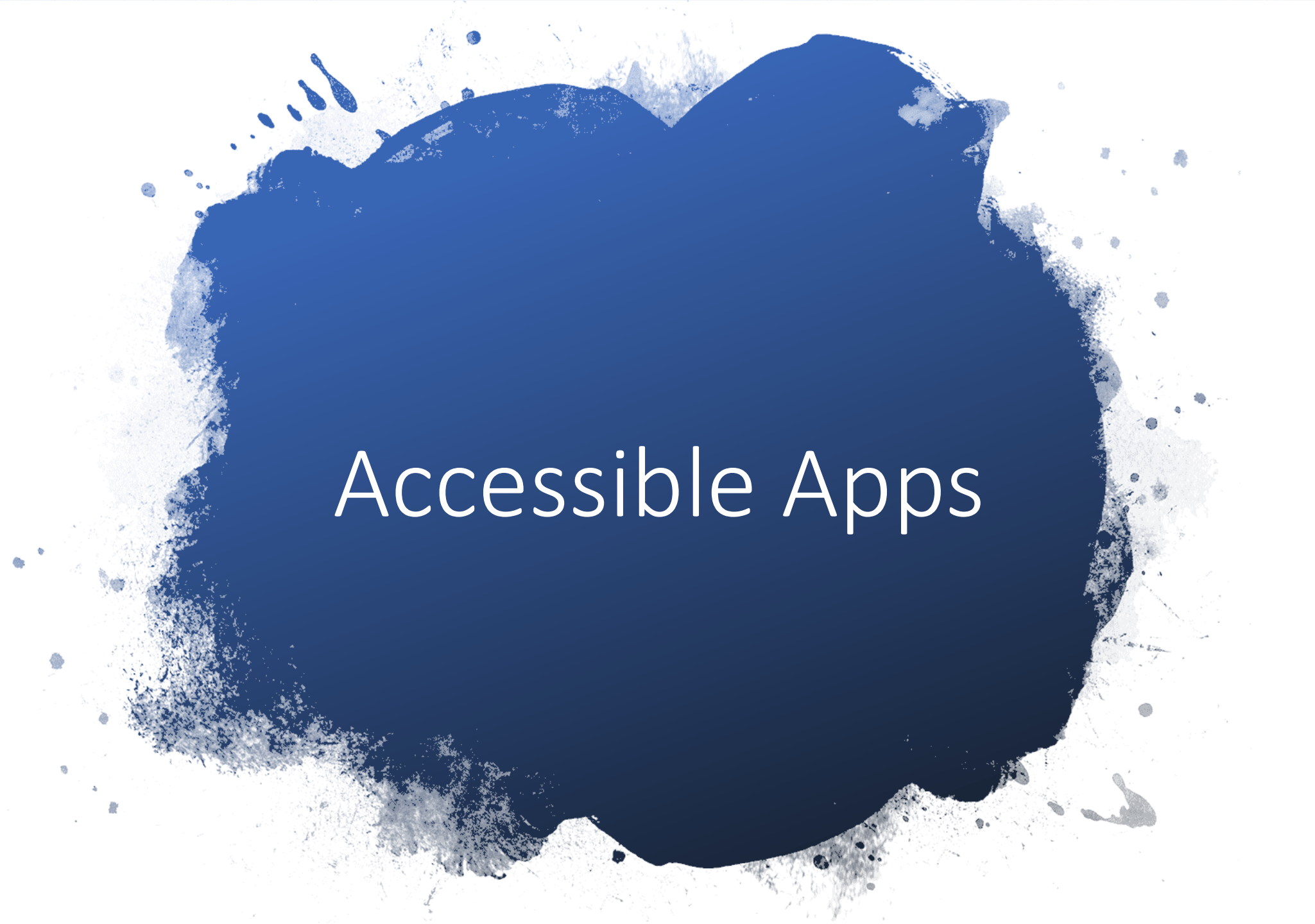
Digital Difficulties / Financial consequences



From government



Limited budget

A large, dark blue, irregular ink blot or splash shape is centered on a white background. The edges of the blot are textured and splattered, with some smaller droplets visible around it. The text "Accessible Apps" is written in a clean, white, sans-serif font across the middle of the blot.

Accessible Apps

Layout

- Simplify the layout
- Bigger / Larger font
- Enough contrast
- Option to toggle dark background / bright background
- Color accents

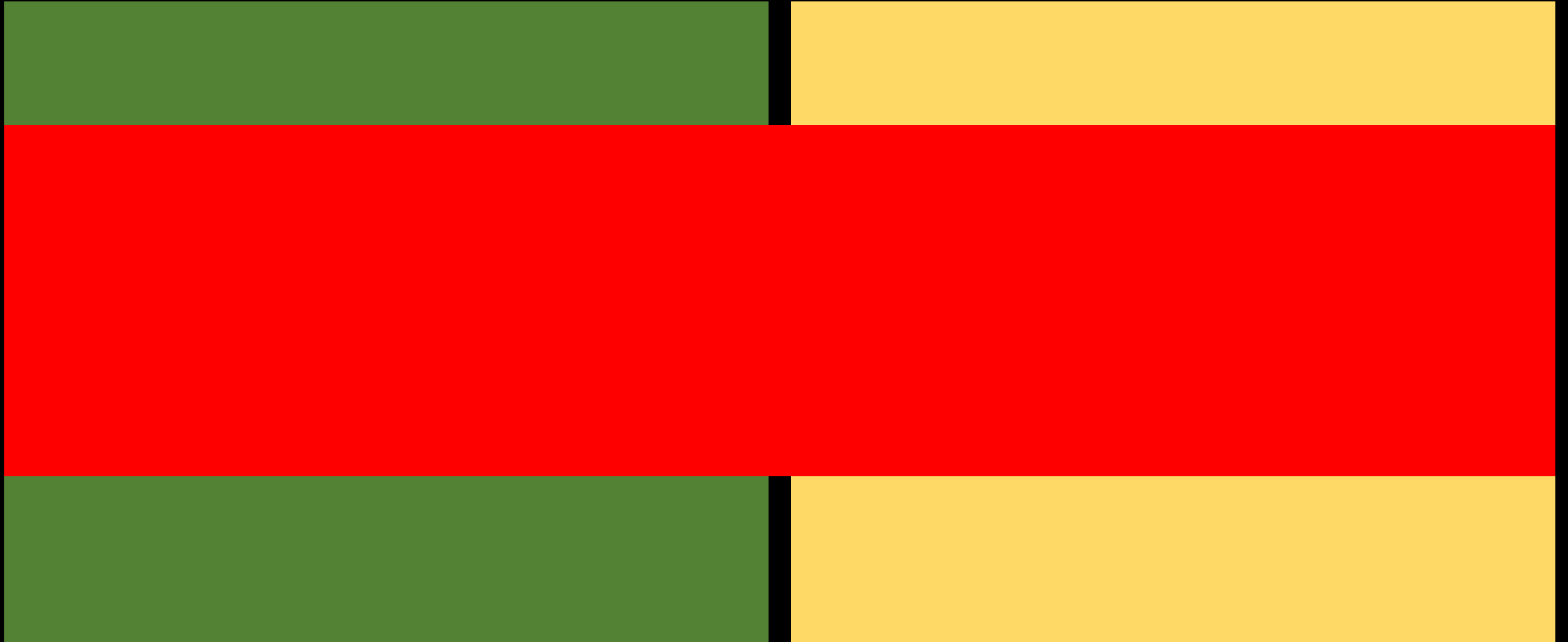
Layout - structure

- Limit content
- Limit options
- 1 topic per screen
- Which screens / controls in which order?

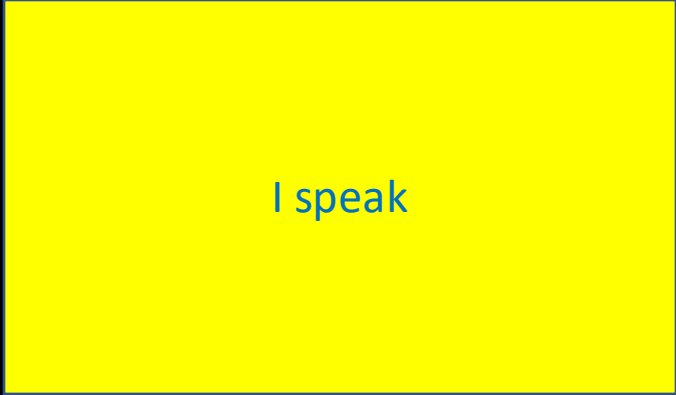
Layout - Direction

- Stack == easier than Grid

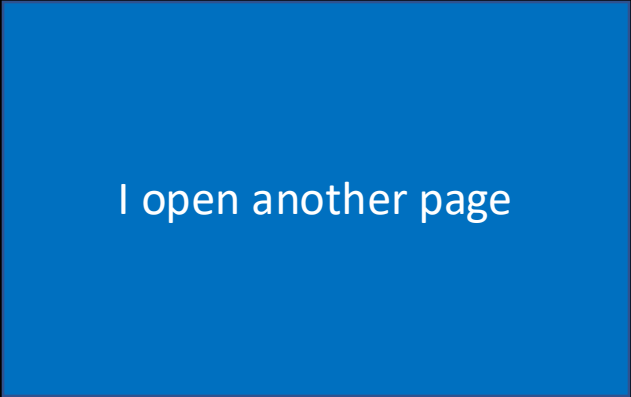
Layout - Colors



Color by function

A solid yellow rectangular box.

I speak

A solid blue rectangular box.

I open another page

A solid gray rectangular box.

I navigate

Important basic colors

Yes

Confirm

No

Cancel

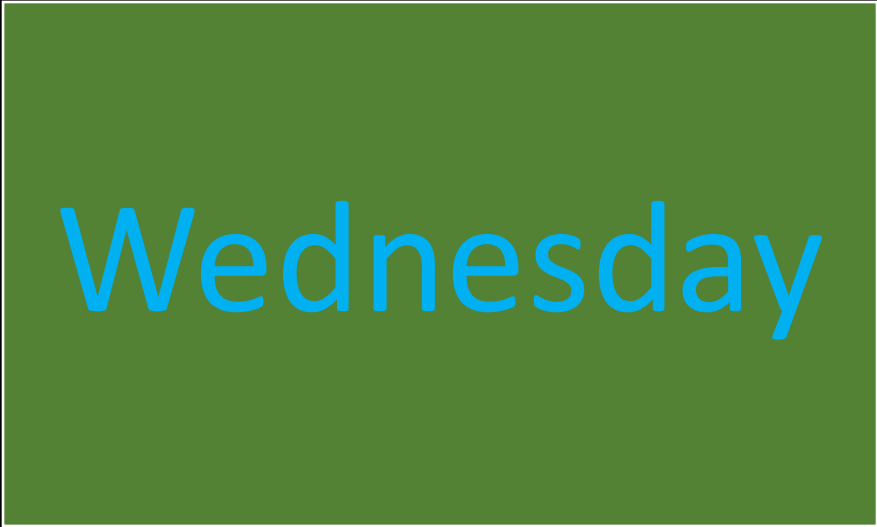
Colors for time indication

A solid yellow rectangular box with a thin black border.

Today

A solid gray rectangular box with a thin black border.

Monday

A solid green rectangular box with a thin black border.

Wednesday



Content

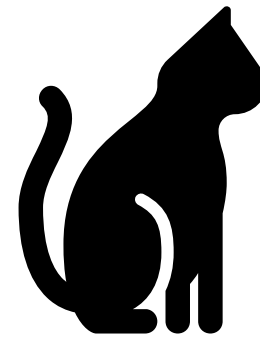
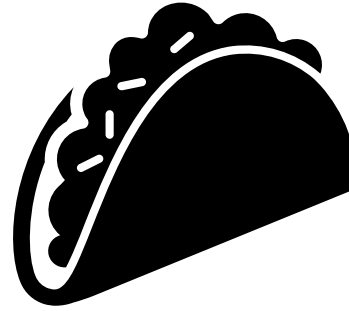


Content - pictures

- What means the following?
- 酢飯
- The answer is:



Content - pictograms





Content – Use of language



APRIL 1914 - THE DASS FOREST COMPANY (BOISCHART & DE BOMCKEN)

Gentse feesten!



De 24^e Gentse Feesten
van 17 tot

Own language

Basic language

Basic

- Fish
- Happy
- Vegetables

Complex

- Salmon; Cod
- Enlightened
- Salad , Coles

Sign language



Partner ecosystem

Choose the best solution for each user!

Spotlight: Accessibility apps and extensions – Recently Added!

36

Visible text in
movies

More deeply in the experiences and deliver new experiences. I'm happy to say that we have

Text to
speech

Intellectual disabilities

Confirmation

On separate controls

Specific buttons

Text To Speech != Narrator / Voice Over



NARRATOR
READS
SEQUENTIAL
EVERY CONTROL



TEXT TO SPEECH
READS THE
CONTROL WHERE
TAPPED ON



NARRATOR IS
GOOD FOR
PEOPLE WITH
VISION LOSS



NARRATOR IS
CONFUSING
FOR PEOPLE
WITH
INTELLECTUAL
DISABILITIES

Content - Limitations

- Password protection



Environmental



4
1

Device

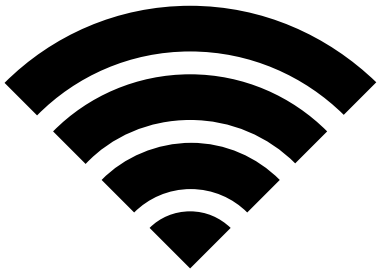
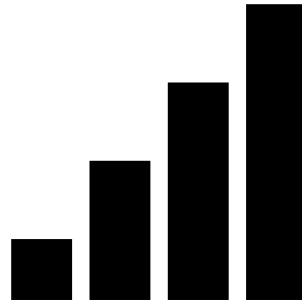
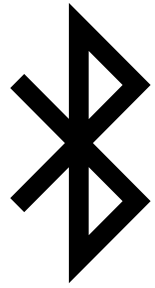
Battery



Deze foto van Onbekende auteur is gelicentieerd onder CC-BY-NC-ND



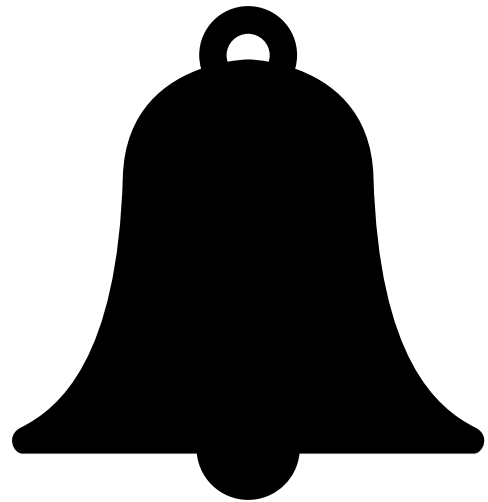
Device info / app info



Connectivity



Only Wi-Fi



Notifications



Make it sexy,
accessible sexy

A good accessible solution



Am I Sexy?
Do you want me?

NO!

I am feeling



Am I still accessible?
Do you want me?

Let me think about this...

For some people it's still accessible...

Things to consider

Some people are color blind.

Some people have low vision.

Keep being aware of Color By Function!



Which apps are
needed?

Clothing



How to Clothe?

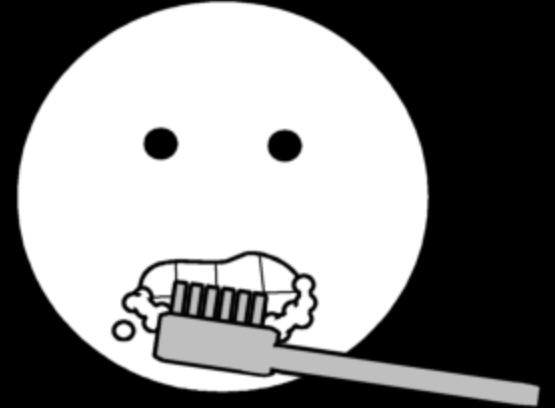
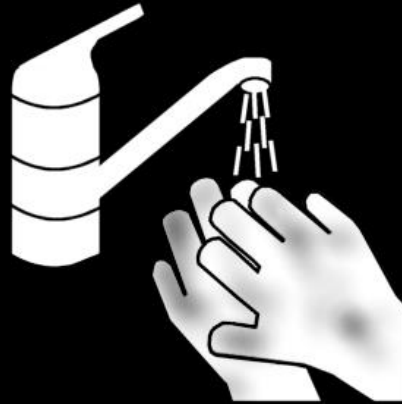
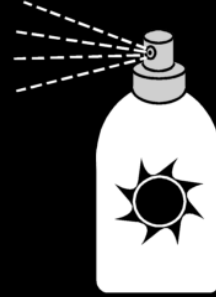
Too hot?

Too cold?

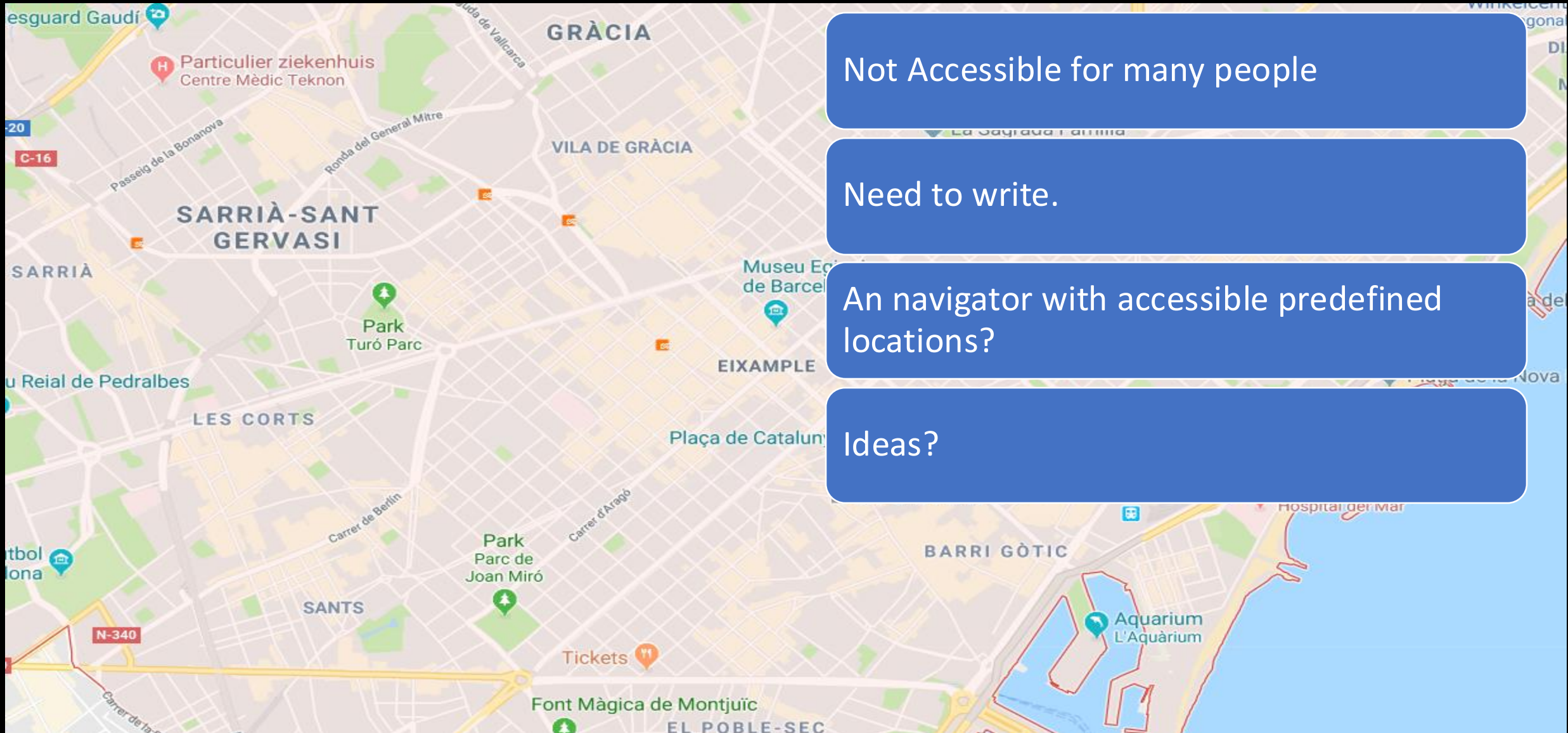
Too casual?

Too sportive?

Hygiene



Navigating



Not Accessible for many people

Need to write.

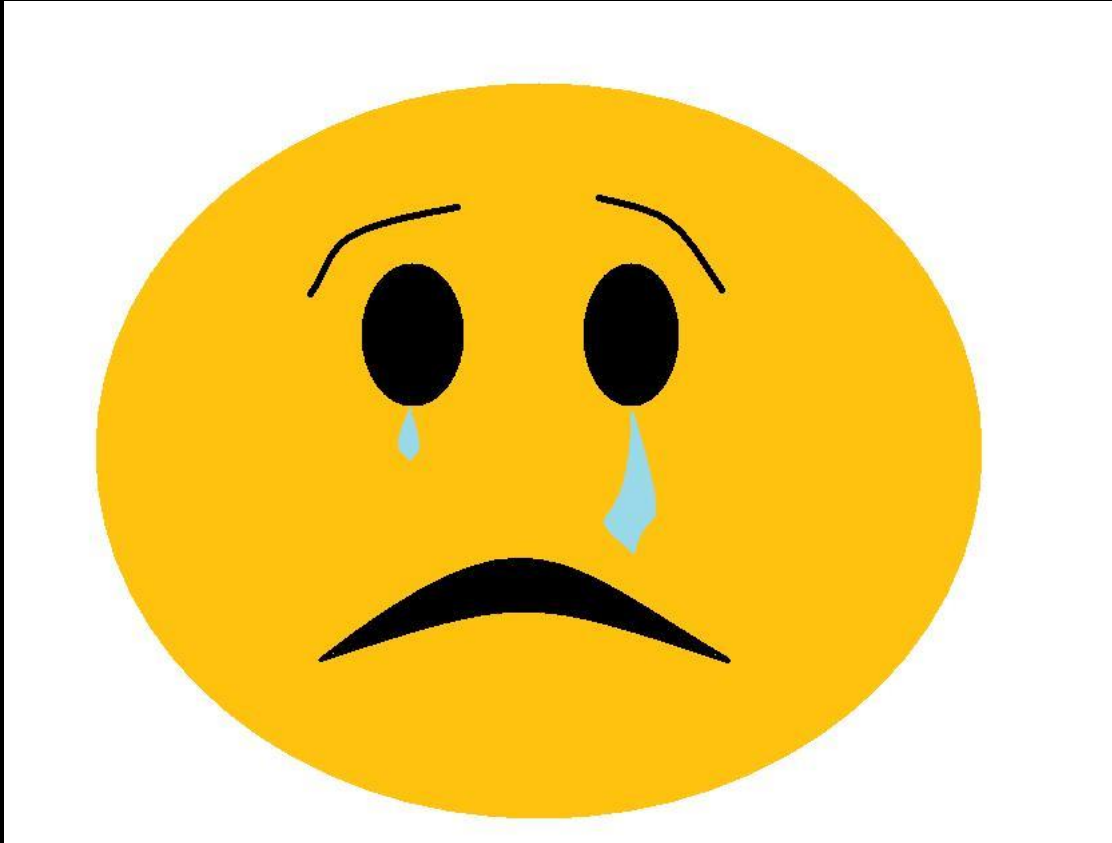
An navigator with accessible predefined locations?

Ideas?

Housekeeping



Mental Health



Communicate about mental health

Digital Assistance

A first step to....

My mental Diary....

Many, many possible...



Conclusion

Thanks || Q&A



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